TITAN EYE EYEWEAR E-COMMERCE

WEEK 7 – SQL QUERY IMPLEMENTATION

1. Perform SELECT, WHERE, ORDER BY, DISTINCT Queries

1.1 Display All Products – SELECT

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       101
Titan Classic Frame
      2699         30           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
```

```
Black
Anti-Glare


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       102
Titan Metal Frame
      3299         15           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Silver
Blue Cut


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
```

```
LENS_TYPE
--------------------------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
UV Protection


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       104
Titan Round Sunglasses
      2799         12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
```

```
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------
Brown
Polarized
```

1.2 Select Specific Columns – SELECT

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       101
Titan Classic Frame
      2699

       102
Titan Metal Frame
      3299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------

       103
Titan Aviator Sunglasses
      2999

       104
Titan Round Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
      2799
```

1.3 Filter Products Using WHERE Condition

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 3000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       101
Titan Classic Frame
      2699         30           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
Anti-Glare


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20          2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
Black
UV Protection


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
       104
```

```
Titan Round Sunglasses
       2799          12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
      PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------
Brown
Polarized
```

```
SQL>
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 2500;

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
----------------------------------------
LENS_TYPE
----------------------------------------
       101
Titan Classic Frame
      2699         30          1

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
----------------------------------------
LENS_TYPE
----------------------------------------
Black
Anti-Glare


PRODUCT_ID
```

```
PRODUCT_NAME
----------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
----------------------------------------
LENS_TYPE
----------------------------------------
       102
Titan Metal Frame
      3299         15          1

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
----------------------------------------
LENS_TYPE
----------------------------------------
Silver
Blue Cut


PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
----------------------------------------
LENS_TYPE
----------------------------------------
       103
Titan Aviator Sunglasses
      2999         20          2
```

1.4 Sort Products in Ascending Order – ORDER BY ASC

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Titan Classic Frame
      2699

       104
Titan Round Sunglasses
      2799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Titan Aviator Sunglasses
      2999

       102
Titan Metal Frame

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      3299
```

1.5 Sort Products in Descending Order – ORDER BY DESC

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
       102
Titan Metal Frame
      3299

       103
Titan Aviator Sunglasses
      2999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------

       104
Titan Round Sunglasses
      2799

       101
Titan Classic Frame

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE
----------
      2699
```

1.6 Display Unique Categories – DISTINCT

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
          1
          2
```

2.1 Search Products by Price Range – BETWEEN

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 2700 AND 3000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
UV Protection


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
```

```
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------
       104
Titan Round Sunglasses
      2799         12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------
Brown
Polarized
```

2.2 Search Products by Category – WHERE

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       101
Titan Classic Frame
      2699         30           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
Anti-Glare
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       102
Titan Metal Frame
      3299         15           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Silver
Blue Cut
```

2.3 Display Available Products – Stock > 0

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM Product
  3  WHERE Stock > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       101
Titan Classic Frame
      2699         30

       102
Titan Metal Frame
      3299         15

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       103
Titan Aviator Sunglasses
      2999         20

       104
Titan Round Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
      2799         12
```

3.1 Display All Customer Details

3.2 Retrieve a Specific Customer's Detail

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 101;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY                           PINCODE
------------------------------ ----------
        101 Arun Kumar
arun@gmail.com
9999999999

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY                           PINCODE
------------------------------ ----------
12 Anna Nagar
Bangalore                       600040
```

3.3 Display Product Information

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Titan Classic Frame
      2699

       102
Titan Metal Frame
      3299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Titan Aviator Sunglasses
      2999

       104
Titan Round Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      2799
```

4. Apply Filtering Conditions

4.1 Filter Using AND Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 3000
  4  AND Stock > 0;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       101
Titan Classic Frame
      2699         30          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- ----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
Anti-Glare
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
Black
UV Protection
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
       104
Titan Round Sunglasses
      2799         12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------
Brown
Polarized
```

4.2 Filter Using OR Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 1
  4  OR Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       101
Titan Classic Frame
      2699         30           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
Anti-Glare
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       102
Titan Metal Frame
      3299         15           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Silver
Blue Cut
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------
Black
UV Protection
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------
       104
Titan Round Sunglasses
      2799         12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------
Brown
Polarized
```

4.3 Filter Using IN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID IN (1, 2);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       101
Titan Classic Frame
      2699         30           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
Anti-Glare


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
```

```
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       102
Titan Metal Frame
      3299         15           1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Silver
Blue Cut


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
```

```
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
UV Protection


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       104
Titan Round Sunglasses
      2799         12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Brown
Polarized
```

4.4 Filter Using BETWEEN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 2700 AND 3000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
       103
Titan Aviator Sunglasses
      2999         20           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------------------------------------
Black
UV Protection


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
```

5.1 Display Product Price Details

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
       102
Titan Metal Frame
      3299

       103
Titan Aviator Sunglasses
      2999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------

       104
Titan Round Sunglasses
      2799

       101
Titan Classic Frame

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
      2699
```

5.2 Display Products in Each Category

```
SQL> SELECT Category_ID, COUNT(*) AS Total_Products
  2  FROM Product
  3  GROUP BY Category_ID
  4  ORDER BY Category_ID;

CATEGORY_ID TOTAL_PRODUCTS
----------- --------------
          1              2
          2              2

SQL>
```

5.3 Display Available Stock Details

```
SQL> SELECT Product_ID, Product_Name, Stock
  2  FROM Product
  3  WHERE Stock > 0
  4  ORDER BY Stock DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     STOCK
----------
       101
Titan Classic Frame
        30

       103
Titan Aviator Sunglasses
        20

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     STOCK
----------

       102
Titan Metal Frame
        15

       104
Titan Round Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
     STOCK
----------
        12
```

```
Titan Round Sunglasses
       2799          12           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
      PRICE      STOCK CATEGORY_ID
---------- ---------- -----------
FRAME_COLOR
--------------------------------------------------
LENS_TYPE
--------------------------------------------------
Brown
Polarized
```